Operationalizing Verification and Validation for Text Corpora: A Parameter-Free Information-Theoretic Framework

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***Abstract*—Corpora underpin empirical work in computational linguistics, yet the corpus itself is rarely subjected to the systematic verification and validation (V&V) that comparable engineering disciplines require. We present a framework that makes corpus V&V operational and measurable. First, we recast the verification/validation distinction as a falsifiable prediction about which measurement instrument detects which class of defect, rather than as a descriptive taxonomy. Second, we introduce Reciprocal Good–Turing (RGT) smoothing, a parameter-free estimator for divergence between two corpora: the reserved probability mass is set by the Good–Turing singleton estimate and redistributed over unseen n-grams in proportion to the comparison partner's distribution. RGT removes the free smoothing parameter and self-adapts across n-gram orders and across languages, requiring no user-selected corpus, language, or n-gram-specific smoothing parameter. We use 'parameter-free' to mean that no user-selected smoothing parameter such as k is required; the smoothing mass is determined from the observed corpus, with fixed boundary rules for degenerate cases. Third, we evaluate detection against controlled ground truth on *six* corpora spanning five languages and two scripts; including the King James Bible, Spanish, Arabic, Igbo, and Nigerian Pidgin, injecting five defect classes at known rates, for 2,400 corpus-level and 360 document-level measurements. Across every language, RGT is monotonic in defect rate for content defects and yields large within-corpus effect sizes (mean $d = 8.3$ for encoding corruption, $d = 5.3$ for foreign-text contamination), whereas a frequency-profile baseline representative of established practice moves in the *wrong direction* under light encoding corruption in all six corpora (d from -2.0 to -9.3). The framework's central prediction holds in every language: structural defects are weak at the corpus level (mean $d \leq 0.8$) but are detected perfectly at the document level (ROC-AUC $= 1.000 \pm 0.000$ in all six), while content defects produce substantially larger corpus-level effects, a separation of at least $6.7\times$ in effect size.**

***Index Terms*—corpus quality, verification and validation, Jensen–Shannon divergence, Good–Turing estimation, smoothing, defect injection, language resources**

I. Introduction

Empirical computational linguistics rests on corpora, but the corpus is seldom treated as an engineered artifact whose fitness for purpose must be demonstrated. Kilgarriff's observation that corpus linguistics harbours a methodological void at its centre [1] remains apposite: the object at the heart of the research is the object least often subjected to quality control.

Recent large-scale audits have made the cost of this omission concrete. A manual audit of 205 language-specific corpora drawn from five major public multilingual datasets found that at least fifteen contained no usable text at all, that a substantial fraction consisted of less than half acceptable-quality sentences and that corpora scores were mislabelled or carried nonstandard language codes [2]. Duplication in training corpora has separately been shown to degrade downstream language models [3]. These are not exotic failures; they are precisely the defects that a routine quality-management process would detect.

Software and system engineering formalizes such a process as *verification and validation* [4], [5]. Verification asks whether the artifact was built correctly against its specification; validation asks whether the right artifact was built for the user's needs. Applied to corpora, verification concerns structure: empty or truncated documents, duplicates, inconsistent markup, mismatched document counts across a parallel pair; while validation concerns content: whether the text is in the language claimed, whether it is representative, whether annotations are correct.

Making this operational requires answering two questions that the literature leaves open. *Which instrument detects which defect?* A taxonomy that classifies defects but does not tell a practitioner what to measure is of limited use. *And how do we know a proposed measure detects anything?* A measure is commonly demonstrated by computing it between two well-known corpora, obtaining a small number, computing it between two dissimilar corpora, obtaining a larger number, and observing that the ordering is intuitive. This establishes that the measure ranks obviously different texts above obviously similar ones. It does not establish detection, because no defect was present to detect and no correct answer was available for comparison.

We address both questions. For the second we construct ground truth by *controlled defect injection*: starting from clean corpora, we inject defects of known type at known rates, so the correct answer is known by construction. This enables dose–response analysis, effect-size estimation, statistical testing, and comparison with a baseline.

Our contributions are:

1) **An operational V&V framework** that maps each defect class to the instrument that detects it, stated as a falsifiable prediction and supported experimentally (Sections III and VI).
2) **Reciprocal Good–Turing smoothing**, an estimator requiring no user-selected corpus-, language-, or n-gram-specific smoothing parameter, for corpus divergence that derives its smoothing mass from the data and redistributes it using the comparison partner rather than uniformly (Section IV-C). It removes a free parameter whose choice changes the measured encoding-corruption effect size by 2.5× over the tested add-k range.
3) **A controlled defect-injection benchmark** for corpus V&V, comprising 2,400 corpus-level and 360 document-level measurements over five defect classes and eight injection rates (Sections V–VI).

II. Related Work

A. Corpus comparison measures

Kilgarriff and Rose surveyed measures for corpus similarity and homogeneity, recommending a χ^2 statistic computed over word-frequency profiles [1], [6]. Frequency-profile comparison remains the established informal instrument for judging whether two corpora resemble one another, and we adopt a normalized form of it as our baseline for exactly that reason.

B. Compression, entropy, and divergence

Compression provides a practical estimator of cross-entropy, and PPM-family models have been used to estimate the entropy of natural language [7], [8]. The connection to divergence is direct: the excess codelength incurred when coding one corpus with a model trained on another is the Kullback–Leibler divergence between them [9]. The Jensen–Shannon divergence [10] provides a symmetric, bounded alternative whose square root satisfies the triangle inequality [11].

C. Smoothing

To obtain finite divergence over the union of the two vocabularies, nonzero probability estimates are required, motivating smoothing. Add-k (Lidstone) and related schemes are surveyed in [12]. Good–Turing estimation [13], [14] instead derives the mass reserved for unseen events from the frequency-of-frequencies of the observed sample. In information retrieval, document language models are commonly smoothed toward a background collection model [15]. Our estimator (Section IV-C) combines the Good–Turing mass estimate with a redistribution conditioned on the comparison partner; Section IV-D states the relationship precisely.

D. Corpus documentation and auditing

Datasheets [16] and data statements [17] standardize what should be *recorded* about a dataset; audits of C4 [18] and of web-crawled multilingual corpora [2] report what is actually *found* in them. These are complementary to our concern: they prescribe documentation and report manual findings, whereas we ask which automatic measurement reliably detects a given defect and demonstrate it against ground truth across languages.

III. An Operational V&V Framework

We take verification to concern the *structure* of a corpus and validation its *content*. The substantive claim is that this is not bookkeeping but a statement about where the evidence for a defect resides.

A structural defect perturbs the arrangement of documents while leaving the aggregate distribution of linguistic material largely intact. Replacing one document with a copy of another changes no token's corpus-wide frequency by more than a small amount; truncating a few documents removes tokens roughly in proportion to their existing frequencies. The aggregate n-gram distribution barely moves, and the evidence lives in relations *between* documents.

A content defect changes the linguistic material itself. Substituting a foreign-language document introduces character sequences absent from the reference; corrupting an encoding introduces symbols outside the script. The aggregate distribution moves, and comparison against a trusted reference is informative.

This yields Table I and the following prediction, which Section VI tests directly in every language:

> *The reported experiments support the framework's prediction that corpus-level divergence is sensitive to content defects while document-level structural detectors identify duplication and truncation across all six corpora.*

TABLE I
Defect classes, V&V category, and predicted instrument.

Defect	Category	Evidence in	Instrument
Duplication	Verif.	document relations	doc-level
Truncation	Verif.	document lengths	doc-level
Boilerplate	Verif.	repeated spans	doc-level
Foreign text	Valid.	aggregate content	corpus-level
Encoding corr.	Valid.	aggregate content	corpus-level

A further consequence is that the n-gram *type* is not a free choice. Some defects operate below the word level. Script substitution and mojibake can corrupt the character sequence itself, making word-level statistics unreliable because those statistics presuppose successful tokenization. We use word unigrams for structural defects and character trigrams for content defects.

IV. A Measure for Corpus Comparison

A. Requirements

The task imposes four requirements: **(R1) symmetry**, since neither corpus is privileged; **(R2) boundedness**, so magnitudes are interpretable across corpus pairs of different sizes; **(R3) definition over the union of the two vocabularies**, because n-grams present in one corpus and absent from the other

are among the most diagnostic evidence and must not be discarded; and **(R4) freedom from arbitrary tuning**, since a quality-control instrument whose verdict depends on an analyst's parameter choice is of limited evidential value.

B. Divergence

We use the Jensen–Shannon divergence [10] between the n-gram distributions P and Q of the two corpora: M is the average distribution of P and Q, Kullback–Leibler divergence $D_{\mathrm{KL}}(P\|M)$ measures how different P is from M;

$$\begin{aligned} \mathrm{JSD}(P,Q) &= \tfrac{1}{2}D_{\mathrm{KL}}(P\|M) + \tfrac{1}{2}D_{\mathrm{KL}}(Q\|M), \\ M &= \tfrac{1}{2}(P+Q), \end{aligned} \tag{1}$$

which satisfies R1 and R2: it is symmetric and bounded, $0 \leq \mathrm{JSD} \leq 1$ bit, equalling zero exactly when $P = Q$. It retains the compression interpretation, since $D_{\mathrm{KL}}(P\|Q)$ is the expected excess codelength in bits per n-gram incurred by coding one corpus with a model trained on the other [9]. Requirements R3 and R4 concern how P and Q are estimated, which is where the difficulty lies.

C. Reciprocal Good–Turing smoothing

Because $\log 0$ is undefined, a divergence over the vocabulary union requires every n-gram to receive nonzero probability under both corpora. Add-k smoothing supplies this but has two properties unsuitable here. It requires choosing k, for which this setting offers no principled value (violating R4). And it distributes the reserved mass *uniformly*: an n-gram absent from corpus A but frequent in B receives the same probability under A as one absent from A and vanishingly rare in B, although the two carry very different evidential weight. A further difficulty is that the effective smoothing strength of add-k is $k|V|/(N + k|V|)$, which depends on vocabulary size; since $|V|$ differs by orders of magnitude between word unigrams and character trigrams, and between languages, a single k imposes materially different smoothing across conditions.

We therefore estimate the reserved mass from the data and redistribute it using the comparison partner. Let corpora A and B have counts C_A, C_B, token totals N_A, N_B, vocabularies V_A, V_B, and union $V = V_A \cup V_B$.

Mass. Reserve for A the Good–Turing estimate of the probability that the next token belongs to a previously unseen type [13], [14],

$$\theta_A = n_1(A)/N_A, \tag{2}$$

where $n_1(A)$ is the number of types occurring exactly once in A. This is a property of the corpus, not a tuning choice.

Redistribution. Let $U_A = V \setminus V_A$ be the n-grams B has and A lacks. Distribute A's reserved mass over U_A in proportion to the partner's distribution there and symmetrically for B:

$$P_A(g) = \begin{cases} (1-\theta_A)\,\dfrac{C_A(g)}{N_A}, & g \in V_A, \\[2ex] \theta_A\,\dfrac{C_B(g)/N_B}{\sum_{g' \in U_A} C_B(g')/N_B}, & g \in U_A, \end{cases} \tag{3}$$

If $U_A = \varnothing$, no unseen support exists on which to redistribute the reserved mass, and the implementation therefore uses the unsmoothed empirical distribution $P_A(g) = C_A(g)/N_A$.

Both are proper distributions over V, since the two branches contribute $(1-\theta)$ and θ respectively. We floor θ at $1/(N+1)$ to keep probabilities strictly positive when a corpus has no singletons, and cap it at 0.5 so observed events always retain the majority of the mass.

The construction has a natural reading. The null model is that the two corpora are samples of the same process; where A provides no evidence, the least-committal assumption is that A resembles B. The divergence then measures departure from that null rather than an artifact of where mass was arbitrarily placed.

D. Relationship to prior work

Good–Turing estimation [13], [14] and smoothing toward a background distribution [15] are both long established. The contribution is their composition for the two-corpus divergence setting: using the Good–Turing mass estimate *and* conditioning its redistribution on the comparison partner rather than on a fixed background or a uniform prior. This removes the free parameter, makes effective smoothing self-adapting across n-gram orders and languages, and; as Section VI-A shows, matches the better part of the add-k range with no parameter set.

E. Properties

We verified the following empirically. Over all pairs and triples drawn from the six corpora of Table II, RGT is symmetric and satisfies the identity of indiscernibles exactly, and all divergences lie in $[0, 1]$.

Metricity requires a caveat we state explicitly. Under fixed smoothing, $\sqrt{\mathrm{JSD}}$ is a true metric [11]. Under RGT the estimated distributions are *pair-dependent*: the P_A used for $d(A, B)$ differs from that used for $d(A, C)$, so the guarantee does not transfer. Empirically $\sqrt{\mathrm{JSD}}$ satisfied the triangle inequality in all triples formed from our corpora and in $9{,}000$ triples of corpora sampled at sizes from 20 to 20,000 tokens; a directed search produces violations only when every corpus in a triple is a handful of tokens in which nearly all types are singletons, driving θ to its cap. On the corpora studied here θ ranges from 0.005 to 0.18 (Table II), far from that regime. Where formal metricity is required, fixed-k smoothing should be preferred; for the detection task studied here, RGT provides larger or comparable effect sizes without a user-selected smoothing parameter..

F. Baseline

As a baseline representing established practice we use the mean absolute difference of relative frequencies over V:

$$B(C_1, C_2) = \frac{1}{|V|} \sum_{g \in V} \left| \frac{C_1(g)}{N_1} - \frac{C_2(g)}{N_2} \right|. \tag{4}$$

V. Experimental Design

A. Corpus substrate

We use six public-domain or openly-licensed corpora spanning five languages and two scripts (Table II). **English** concatenates six public-domain novels. **Bible** is the King James Version, an archaic English register. **Spanish** is *Don Quijote*. **Arabic** is the Qur'ān, our non-Latin (Arabic-script) condition. **Igbo** and **Nigerian Pidgin** are drawn from the AfriBERTa corpus [19], predominantly BBC News; Igbo is a Niger–Congo language written in Latin script with subdot and tonal diacritics, and Nigerian Pidgin is an English-lexified creole. Pidgin is included deliberately as a *same-script, related-lexicon* stress case for foreign-text detection. Project Gutenberg licence boilerplate was stripped and its absence verified. The corpora span a wide range of the smoothing statistic θ (0.005–0.18), exercising the smoothing estimator across conditions.

TABLE II
CORPUS SUBSTRATE: SIX CORPORA, FIVE LANGUAGES, TWO SCRIPTS. θ IS THE DATA-DERIVED WORD-LEVEL SMOOTHING STRENGTH.

Corpus	Language / note	Tokens	Types	θ
English	English (6 novels)	844,127	26,660	0.011
Bible	English (archaic)	854,044	12,744	0.005
Spanish	Spanish	381,256	22,960	0.029
Arabic	Arabic (script)	77,429	21,117	0.180
Igbo	Igbo (diacritics)	498,421	29,591	0.035
Pidgin	Nigerian Pidgin	446,759	18,827	0.022

B. Protocol

For each randomization seed, the corpus documents were independently shuffled and divided into trusted REFERENCE and TEST halves; the same seed-specific split was retained across all injection rates for that randomization. Because both halves originate in the same corpus, their divergence at zero injection provides the null distribution *for that corpus*. A defect is injected into the TEST half at rate $r \in \{0, .05, .10, .15, .20, .30, .40, .50\}$, and we record divergence under RGT, add-k, and the frequency baseline. Because injection is programmatic, the correct answer is known by construction.

Five defect classes are implemented: **duplicate** (documents replaced by copies of others), **truncate** (documents cut to five tokens), **boilerplate** (a repeated promotional string appended), **foreign** (documents replaced by documents from another corpus), and **encoding** (for encoding corruption, a fraction r of TEST documents was selected, and each non-whitespace character within a selected document was independently replaced with probability 0.15 by a symbol drawn from a fixed set of mojibake-like characters). For foreign contamination each corpus is contaminated by a script-different partner where possible; the exception is Pidgin, contaminated with English (same script, shared lexicon) as the hardest case.

Six corpora $\times$ 10 seeds give $n = 60$ independent randomizations conditional on the corpus per (defect, rate) cell; with five defects and eight rates this yields 2,400 corpus-level measurements, plus 360 document-level runs.

C. Reporting effect sizes: within corpus, not pooled

Effect sizes must be computed *within* each corpus and then averaged, not by pooling measurements across corpora. Because corpora differ in their baseline ($r = 0$) divergence, pooling inflates the standard deviation and attenuates the effect size. For foreign contamination, for instance, the naive pooled effect size at $r = 0.05$ is $d = 0.74$, whereas every individual corpus shows d between 3.5 and 7.6. We therefore report within-corpus effect sizes throughout.

D. Document-level detectors

For structural defects we score each document for suspicion. Duplication is scored as $1 - \min_{j \neq i} \sqrt{\mathrm{JSD}}(d_i, d_j)$, so an exact copy scores 1; ground truth labels *both* members of a duplicate pair. Truncation is scored by inverse length. We report ROC-AUC against the injected labels.

VI. Results

A. Smoothing

The smoothing statistic θ is self-adapting (Table II): across corpora it ranges from 0.005 (Bible, lexically repetitive) to 0.18 (Arabic, which has many singleton types in this corpus), and within a corpus it falls by roughly two orders of magnitude from word unigrams to character trigrams. A single fixed k cannot reproduce this data-dependent variation across corpora and n-gram representations.

The choice of k is consequential. Fig. 1 sweeps k over three orders of magnitude at the hardest operating point ($r = 0.05$), reporting the within-corpus effect size averaged over the six corpora. For encoding corruption, it ranges from 4.6 at $k = 5$ to 11.6 at $k = 0.001$, a 2.5$\times$ difference driven entirely by a parameter choice; RGT attains 8.3, within the strong part of that range. For foreign contamination add-k ranges from 4.8 to 6.7 and RGT attains 5.3. RGT thus matches good fixed settings without any choice being made, and it does so simultaneously for both defect types and all six languages.

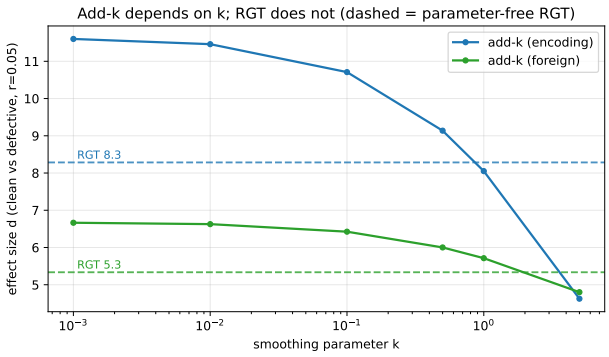


Fig. 1. Within-corpus effect size at $r = 0.05$ (mean over six corpora) as a function of the add-k parameter (solid), with RGT as dashed reference lines. RGT matches the strong part of the k range for both defect classes with no parameter set.

B. Dose–response and the content/structural split

Fig. 2 shows mean divergence against injection rate, averaged over the six corpora. Content defects (foreign, encoding) rise steeply and separate cleanly from the $r = 0$ baseline distribution; the three structural defects rise only shallowly, as Section III predicts. Table III gives the within-corpus effect sizes at $r = 0.05$: content defects reach mean $d = 8.3$ and 5.3, structural defects at most 0.8, a separation of at least $6.7\times$.

To get the effect sizes, population standard deviation is used:

$$d_{c,r} = \frac{\bar{D}_{c,r} - \bar{D}_{c,0}}{\sqrt{(s^2_{c,r} + s^2_{c,0})/2}},$$

where $D_{c,r}$ denotes the divergence values for corpus c at injection rate r, and $D_{c,0}$ denotes the corresponding zero-injection values. The reported table values are the arithmetic mean of $d_{c,r}$ over the six corpora.

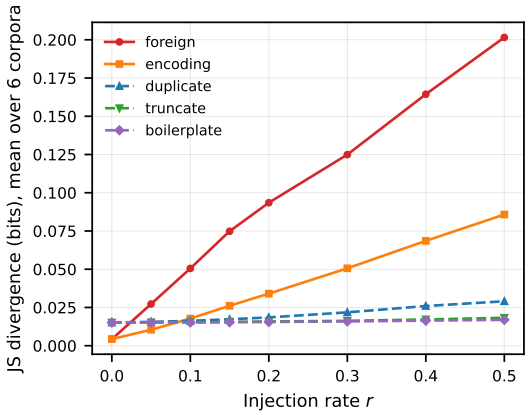


Fig. 2. Dose–response under RGT, mean over six corpora. Content defects (solid) rise steeply; structural defects (dashed) stay near the null.

TABLE III
WITHIN-CORPUS EFFECT SIZE d AT $r = 0.05$ (RGT), MEAN AND RANGE OVER THE SIX CORPORA. CONTENT DEFECTS SHOW SUBSTANTIALLY LARGER EFFECT SIZES THAN STRUCTURAL DEFECTS.

Defect	Class	mean d	range
encoding	content	8.28	$[4.55, 13.88]$
foreign	content	5.34	$[3.49, 7.61]$
duplicate	structural	0.80	$[-0.22, 1.65]$
truncate	structural	0.24	$[0.10, 0.37]$
boilerplate	structural	0.21	$[-0.75, 0.95]$

C. The frequency baseline moves the wrong way, in every language

Table IV contrasts RGT with the frequency baseline on encoding corruption, per corpus, at $r = 0.05$. RGT detects the defect strongly in all six languages (d from 4.6 to 13.9). The baseline has a *negative* effect size in all six (-2.0 to -9.3): under light corruption it reports a more corrupted corpus as more similar to its clean reference. It is non-monotonic in every language. Fig. 3 shows the divergence: RGT rises monotonically in each language while the baseline does not.

The mechanism is instructive. Mojibake substitution alters the corrupted profile by replacing frequent characters with symbols that were rare or absent in the clean corpus. This can move the corrupted profile closer to the reference under an unweighted mean-absolute-frequency criterion, even while the information-theoretic divergence increases. The corrupted test distribution is being altered toward a flatter character-frequency distribution, while the clean reference remains unchanged. Depending on the baseline's averaging over the vocabulary, this can reduce the mean absolute frequency difference even though the test distribution has become less faithful to the reference.Whether a real defect is detected or concealed thus depends on the choice between a frequency heuristic and an information-theoretic quantity, and this holds across languages and scripts.

TABLE IV
ENCODING CORRUPTION AT $r = 0.05$: WITHIN-CORPUS EFFECT SIZE d FOR RGT VS. THE FREQUENCY BASELINE. THE BASELINE IS WRONG-SIGNED IN EVERY LANGUAGE.

Corpus	RGT d	baseline d
English	$+5.86$	-2.69
Bible	$+6.99$	-6.51
Spanish	$+9.54$	-9.30
Arabic	$+4.55$	-7.41
Igbo	$+13.88$	-1.99
Pidgin	$+8.89$	-3.18

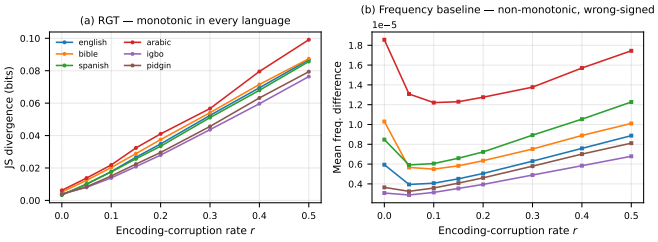


Fig. 3. Encoding corruption across six corpora. (a) RGT rises monotonically in every language. (b) The frequency baseline is non-monotonic and wrong-signed at low corruption in every language.

D. Foreign contamination, including the same-script hard case

Table V reports foreign contamination per corpus. Detection is strong in every case; and critically, the deliberate same-script hard case, Pidgin contaminated with English (no script change, shared lexicon), is still detected at $d = 3.53$. The sensitivity floor of the method is therefore high: it does not rely on a change of script. As expected, the largest signal comes from the largest script/register distance (Arabic contaminated with English, $d = 7.61$). Fig. 4 shows the dose–response per corpus.

TABLE V
FOREIGN CONTAMINATION PER CORPUS (RGT): DIVERGENCE AT r=0 AND r=0.5, AND WITHIN-CORPUS EFFECT SIZE AT r=0.05. PIDGIN←ENGLISH IS THE SAME-SCRIPT, SHARED-LEXICON HARD CASE.

Host	Contaminant	r=0	r=0.5	d@0.05
Arabic	English	0.006	0.689	7.61
Spanish	Arabic	0.003	0.141	7.27
Igbo	Arabic	0.004	0.111	5.06
Bible	Arabic	0.005	0.079	5.04
Pidgin	English	0.004	0.115	3.53
English	Arabic	0.003	0.074	3.49

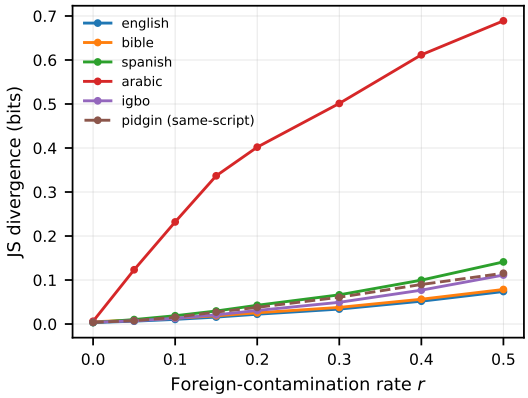


Fig. 4. Foreign-contamination dose–response per corpus (RGT). Pidgin (dashed) is the same-script hard case; it is still clearly detected.

E. The framework's prediction holds in every language

The structural defects of Table III are weak at the corpus level (mean $d \leq 0.8$). Moved to the document level, different instruments detect the same defects without error in *every* corpus (corpus-level: JSD, document-level duplication: nearest-neighbour similarity, document-level truncation: inverse length): ROC-AUC = 1.000 ± 0.000 for both duplication and truncation at every non-zero injection rate tested. The verification/validation distinction thus makes a correct, falsifiable prediction about which instrument succeeds, and it does so language-independently within our experiment, converting a descriptive taxonomy into an operational one.

VII. THREATS TO VALIDITY

Absolute divergence values differ across corpora and genres, so we avoid absolute interpretive thresholds and rely on the within-corpus null the injection protocol supplies. Effect sizes are reported within corpus for the reason given in Section V-C.

The truncation condition is deliberately severe, reducing affected documents to five tokens; consequently, the resulting document-level performance establishes detection under extreme truncation rather than sensitivity to minimally truncated documents.

Injected defects are synthetic. They are drawn from classes reported in real audits [2], [18], but real corpora may exhibit correlated or subtler defects. Synthetic injection buys a known answer at the cost of realism; we consider this the correct trade for establishing detection, and a precondition for studies on naturally occurring defects.

RGT trades formal metricity for adaptivity, as quantified in Section IV-E. For detection this is immaterial, but applications requiring the triangle inequality should use fixed smoothing. Finally, divergence from a reference is not the same as poor quality: a corpus in a new genre *should* diverge from an established reference. Our protocol sidesteps this by comparing a corpus against a clean sample of itself.

VIII. CONCLUSION AND FUTURE WORK

We have treated corpus verification and validation as an engineering process with measurable outcomes, and supplied three components: a framework that maps defect classes to detection instruments; a divergence estimator requiring no user-selected corpus, language, or n-gram-specific smoothing parameter; and a multilingual controlled ground truth against which detection is demonstrated rather than illustrated. Reciprocal Good–Turing smoothing removes a free parameter whose choice alters measured effect size more than twofold, and self-adapts across n-gram orders and across six corpora spanning five languages. The measure is monotonic in defect rate for content defects in every language, whereas a frequency-profile baseline moves the wrong way under light encoding corruption in all six. The verification/validation distinction correctly predicts which instrument detects which defect, content defects at the corpus level, structural defects at the document level with ROC-AUC $= 1.000$, language-independently.

Two limitations frame the future work. First, our substrate, while spanning two scripts (Latin and Arabic), is confined to *space-delimited* writing systems, in which whitespace marks word boundaries. The word-level track therefore does not yet apply to scripts that require prior segmentation; Chinese, Japanese, and Thai, or logographic systems whose very large character inventories would also change the character-level smoothing dynamics; extending the framework to these is the natural next step. Second, the injected defects are synthetic; evaluating the estimator against corpora carrying naturally occurring, human-annotated defects, and extending the injection protocol to parallel corpora so that sentence and, paragraph-alignment pipelines can be treated the same way, remain open.

REPRODUCIBILITY

All corpora are public or openly licensed; the estimator, injection harness, experiment drivers, and analysis scripts are released with fixed random seeds, and every figure and table regenerates from them.

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